OUTPUT SCREENSHOTS

1)Balanced Parenthesis

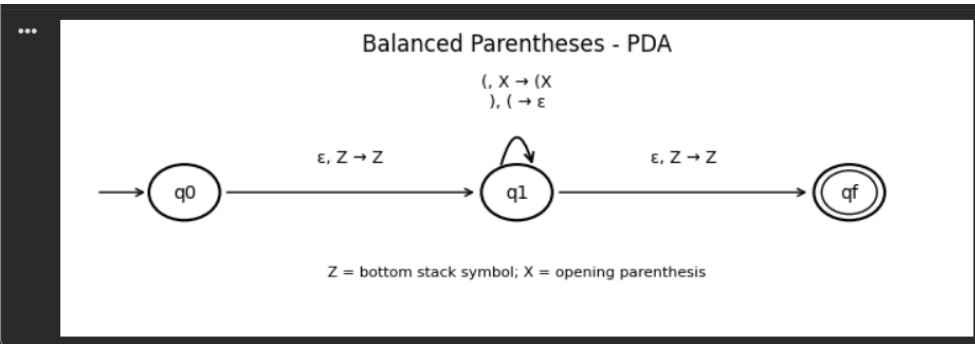
PDA Implementation and Test Cases

```
BALANCED PARENTHESES TEST RESULTS
========================================
() : ACCEPTED
(()) : ACCEPTED
()() : ACCEPTED
(()()) : ACCEPTED
(() : REJECTED
()) : REJECTED
)( : REJECTED
((())) : ACCEPTED

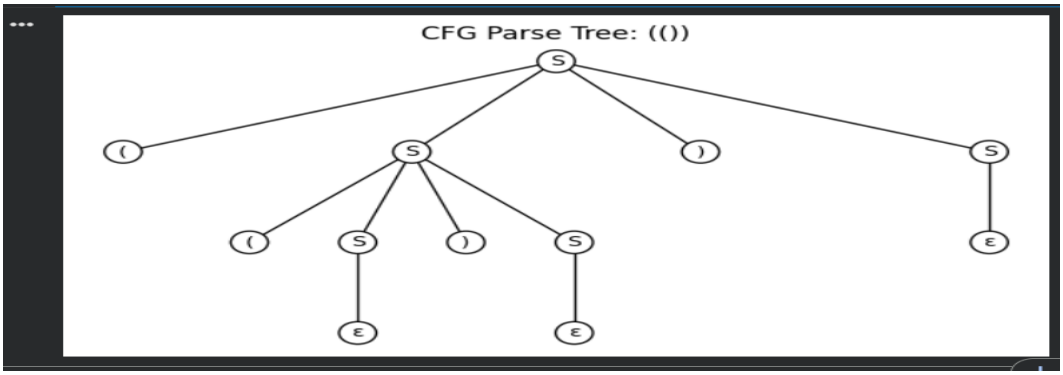
Enter parentheses: )()))()))

Input: ')()))()))'
Initial Stack: ['$']
Invalid closing parenthesis
Final State: qd
Result: REJECTED
'REJECTED'
```

PDA State Diagram



CFG Parse Tree



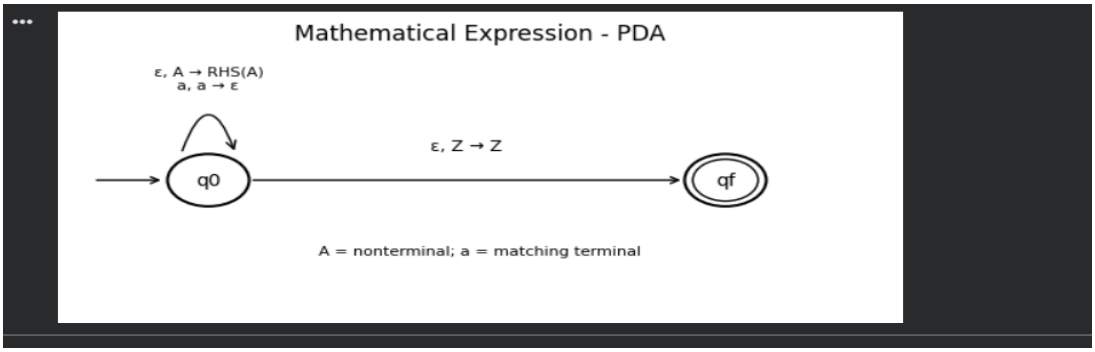
2)Mathematical Expression Verification

PDA Implementation and Test Cases

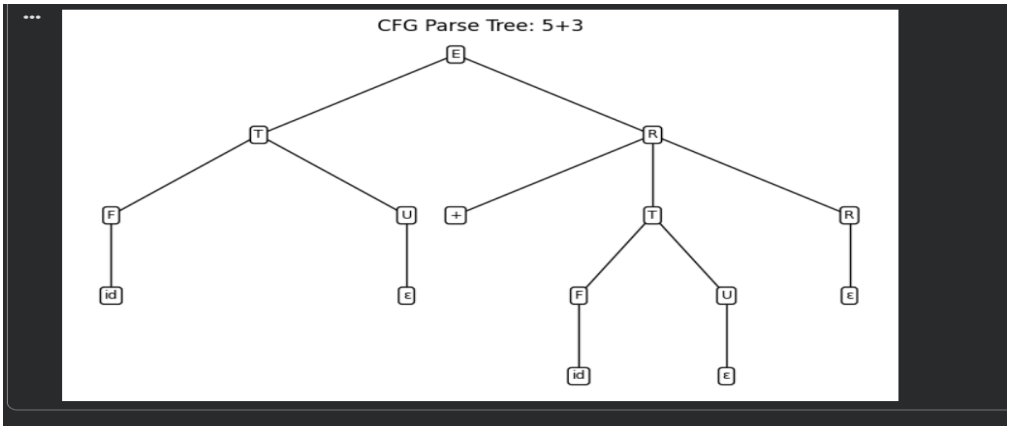
```
Enter mathematical expression: 78+*(56)

Expression: 78+*(56)
Initial Stack: ['$', 'E']
E -> T R
Remaining Stack: ['$', 'R', 'T']
T -> F U
Remaining Stack: ['$', 'R', 'U', 'F']
F -> id
Remaining Stack: ['$', 'R', 'U', 'id']
Matched: id
Remaining Stack: ['$', 'R', 'U']
U -> ε
Remaining Stack: ['$', 'R']
R -> + T R
Remaining Stack: ['$', 'R', 'T', '+']
Matched: +
Remaining Stack: ['$', 'R', 'T']
Invalid production: T *
Result: REJECTED
'REJECTED'
```

PDA State Diagram



CFG Parse Tree



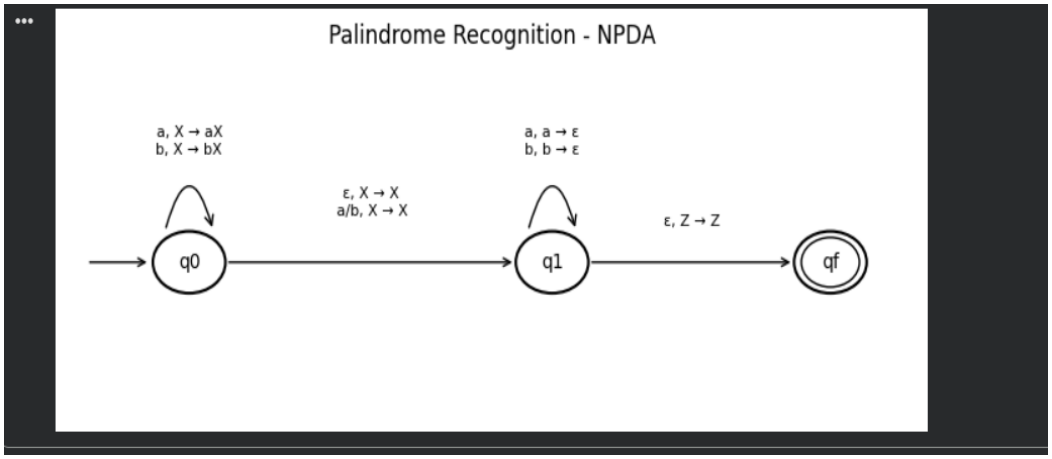
3)Palindrome String Recognition

PDA Implementation and Test Cases

```
PALINDROME RECOGNITION TEST RESULTS
==========================================
'aba' : ACCEPTED
'abba' : ACCEPTED
'a' : ACCEPTED
'bb' : ACCEPTED
'baab' : ACCEPTED
'ab' : REJECTED
'aab' : REJECTED
'baba' : REJECTED
'abcba' : REJECTED
'' : ACCEPTED

Enter a string using a and b: laalaal
Invalid alphabet
Result: REJECTED
'REJECTED'
```

PDA State Diagram



CFG Parse Tree

